
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

Anonymous authors

Anonymous submission

ABSTRACT

Mixture of Experts (MoE) models are now a central mechanism for scaling large language models, but their representational geometry remains less well understood than that of dense networks. Prior superposition work explains how dense models represent more features than dimensions, with feature sparsity and feature importance governing when polysemantic representations emerge. We show that MoEs require a different mechanistic account: in our toy-model setting, feature sparsity and feature importance do not induce the same discontinuous phase changes observed in dense models, whereas network sparsity—the ratio of active experts to total experts—more directly organizes MoE behavior. We introduce metrics for measuring feature capacity and superposition across experts, and find that greater network sparsity yields greater *monosemanticity*. We further define expert specialization in terms of monosemantic feature representation rather than aggregate load balancing, and show that appropriately initialized experts organize around coherent feature combinations. Within the controlled reconstruction setting studied here, these results identify network sparsity as a structural route to more interpretable MoE representations without requiring a corresponding loss of reconstruction performance.

1 INTRODUCTION

Mixture of Experts (MoEs) are a leading architecture for scaling state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), because conditional computation improves computational efficiency while preserving strong performance (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work has refined routing mechanisms, including Expert-Choice routing, and improved training stability and transfer through approaches such as ST-MoE (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical analyses further show that learnable routers can recover latent cluster structure in data, offering an explanation for why experts specialize (Dikkala et al., 2023). Despite this progress, the internal feature representations learned by MoEs remain insufficiently characterized from a mechanistic interpretability perspective.

Recent interpretability-oriented MoE designs make expert behavior more transparent but do not fully explain the feature geometry induced by conditional computation. Yang et al. (2025b) propose MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing through expert activation. These approaches demonstrate that MoE architecture can be shaped toward interpretability, yet they leave open the mechanistic question of how experts represent features, how routing changes superposition, and whether specialization can arise without additional regularization. Mu & Lin (2025) accordingly identify mechanistic interpretability as a major open problem in MoE research.

Superposition is a central obstacle to mechanistic interpretation: models can represent more features than they have dimensions by packing sparse features into shared representational directions. This strategy is capacity-efficient, but it makes individual neurons polysemantic

and thereby weakens the correspondence between a neuron-level measurement and a single interpretable feature.

MoE architectures add a distinct axis to this problem: network sparsity. Unlike dense models, which activate all neurons for every input, MoEs activate only a subset of their total parameters (Shazeer et al., 2017). Dense models therefore exploit feature sparsity by compressing many rarely co-active features into shared directions, whereas MoEs can route inputs to a restricted set of experts and thereby localize representational competition among feature combinations.

We ask three connected questions. First, do MoEs exhibit less superposition than dense models under matched parameter budgets? Second, do MoE experts display the discrete phase changes in feature representation that dense toy models exhibit as feature sparsity and relative feature importance vary? Third, can expert specialization be characterized directly through feature representation, rather than only through aggregate load-balancing statistics?

We study these questions by extending the toy-model framework of Elhage et al. (2022) from dense autoencoders to MoEs. The paper makes three contributions. First, it shows that MoEs do not reproduce the sharp dense-model phase changes; as network sparsity increases, feature representations transition more continuously across sparsity and importance regimes. Second, it demonstrates that MoEs exhibit greater monosemanticity than dense models under equivalent active and total parameter comparisons, with interference confined more locally within experts. Third, it introduces a feature-level definition of expert specialization based on monosemantic representation and expert occupation of feature directions, showing that experts organize around coherent feature combinations rather than arbitrary load balancing alone.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis posits that networks encode concepts as directions in activation space (Park et al., 2024), but the number of interpretable features can exceed the available dimensionality. Elhage et al. (2022) formalized this mismatch as superposition, showing that dense models use non-orthogonal feature packing to increase representational capacity at the cost of polysemanticity. Although superposition can be theoretically efficient (Scherlis et al., 2025), it also motivates post-hoc disentanglement methods, including Sparse Autoencoders, for recovering monosemantic features (Bricken et al., 2023). Recent work has further examined how data correlations shape superposition (Prieto et al., 2025) and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and show that, in this setting, network sparsity is an organizing variable for representational strategy rather than a peripheral architectural detail.

Interpretability of MoEs. Although MoEs are now a standard route to scalable language modeling (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic accounts of their internal representations remain less developed than comparable accounts for dense models. Existing analyses emphasize macroscopic behavior, including routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused approaches induce specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b) or expert-count scaling aligned with vocabulary size (Park et al., 2025). These studies clarify important uses of expert modularity, but they do not isolate the intrinsic feature geometry produced by routing. We address this gap by measuring how experts alter superposition, by showing that MoEs produce more monosemantic representations than dense models in matched toy settings, and by defining specialization at the level of represented features.

3 BACKGROUND

Mechanistic interpretability aims to reverse engineer neural networks by decomposing their internal representations into human-interpretable concepts, often called ‘features’. Under the linear representation view, a hidden state h can be expressed as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ denotes the direction associated with feature i , α_i denotes its activation strength, and F is the set of represented features. In this formulation, feature activation provides an approximate measure of the extent to which the corresponding concept is present in the input.

The *superposition hypothesis* states that models can represent substantially more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Such overcomplete representation requires feature vectors in F to be packed non-orthogonally. When two features interfere, so that $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we describe them as being in *superposition*. This arrangement can be effective when features are sparsely active, as is typical of many text-derived feature distributions, but it reduces interpretability because α_i may contain activations induced by other features rather than by the presence of feature i itself.

Monosemanticity is typically defined at the neuron level: a neuron is monosemantic when its activation corresponds cleanly to a single α_i , meaning that features are aligned with the basis. Orthogonal but non-basis-aligned features can still leave individual neurons *polysemantic*, even when feature-level interference is low. Our analysis therefore targets reduction of superposition rather than explicit enforcement of basis alignment. When we describe models, experts, or features as “more monosemantic,” we use this as shorthand for lower superposition and cleaner feature-level separation.

4 DEMONSTRATING SUPERPOSITION

MoEs are often described as collections of dense subnetworks, with each expert treated as a dense model applied conditionally. That description is architecturally accurate but mechanistically incomplete: it does not determine whether experts learn the same kinds of superposed feature representations as dense models. We therefore begin by directly comparing dense and MoE feature geometry under controlled reconstruction tasks.

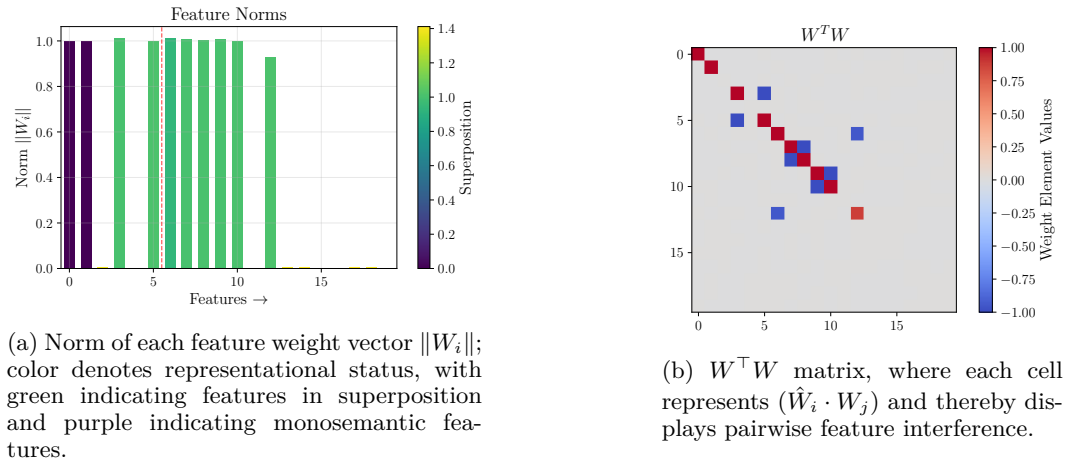


Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, importance $I = 0.7^i$, and uniform feature density $(1 - S) = 0.1$. Color encodes superposition as $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our experimental framework asks how an MoE projects a high-dimensional input vector $x \in \mathbb{R}^n$ into lower-dimensional expert states $h \in \mathbb{R}^m$ and reconstructs the original input. This formulation extends the controlled autoencoder setting of Elhage et al. (2022) to conditional expert architectures.

The input distribution. The input vector x represents activations from an idealized disentangled model in which each dimension x_i corresponds to a distinct independent feature, i.e. a perfectly neuron-aligned and monosemantic representation. We sample $x_i \sim U(0, 1)$ and impose sparsity $S \in [0, 1]$ by setting $P(x_i = 0) = S$. Because features need not contribute equally to reconstruction quality, each feature x_i receives a scalar importance weight I_i in the loss. To isolate phase behavior for a single feature, we additionally scale the final feature by $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. Together, I_i and r control feature signal strength and allow us to measure how expert representations respond to feature magnitude.

Model Architecture. The MoE contains E experts. Each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Each selected expert maps the input to a lower-dimensional hidden state $h^e = W^e x$ and produces the reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The model output is the weighted aggregate over selected experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where w_e denotes the renormalized gating weight.

Models are trained with an importance-weighted L2 reconstruction objective, $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, we add the standard auxiliary load-balancing loss (Fedus et al., 2022) to prevent expert collapse, defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the number of experts, f_e is the fraction of batch samples routed to expert e , P_e is the average gating probability assigned to expert e over the batch, and $\alpha = 0.01$ sets the penalty strength. We omit auxiliary load balancing in Section 4 so that the comparison isolates the architectural bias of MoEs with respect to superposition.

4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we measure two complementary quantities: *representation strength* and *interference*. Representation strength for feature i in expert e is the norm $\|W_i^e\|$, which approaches one when the feature is fully represented by that expert and approaches zero when it is not learned. Interference is visualized through the Gram matrix $W^\top W$, whose off-diagonal entries quantify pairwise interactions between feature directions within an expert.

Figures 1a and 2a show that, under equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$), the dense model and MoE represent a comparable number of features (10 versus 8) with similar feature norms. The crucial difference is not total feature count but interference structure. Although individual MoE experts can exhibit local superposition, as in Expert 0 of Fig 2b, the global interference pattern is partitioned by routing. In the dense model, any represented feature can interfere with any other feature (Figure 1b); in the MoE, features assigned to different experts have zero cross-expert interference, yielding a block-diagonal structure. This comparison establishes that MoEs allocate capacity by partitioning the feature space and thereby reducing the global scope of interference.

Expert Feature Dimensionality. To characterize how MoEs allocate limited representational capacity relative to dense models, we measure feature dimensionality: the “fraction of a dimension” assigned to a given feature (Elhage et al., 2022). For feature i in expert e , dimensionality is defined as

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^\top W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

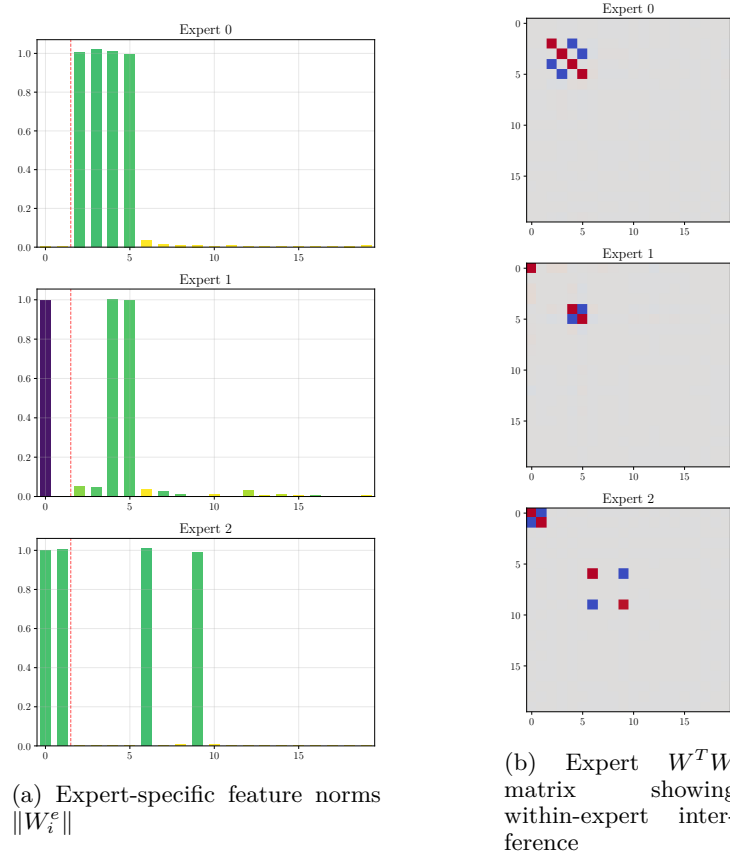


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, with importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero, indicating that the feature is not learned, and one, indicating monosemantic representation. The aggregate MoE capacity is therefore $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. Efficient feature packing occurs when total feature dimensionality matches the available embedding dimensionality, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For an MoE, the corresponding relation is $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, showing that MoEs preserve the parameter efficiency of dense feature packing while changing how interference is distributed across experts.

Features per Dimension & Network Sparsity. Because both dense and MoE models efficiently pack features, the decisive comparison is the number of represented features per available dimension. This statistic directly measures superposition: values above one indicate that the model represents more features than dimensions and therefore relies on superposed representations. For an MoE, we define features per dimension as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

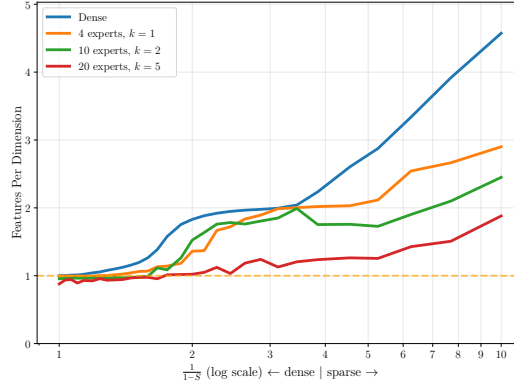


Figure 3: Features per dimension as a function of inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition; superposition decreases as the number of experts increases: 4 experts with $m = 5$, $k = 1$ (orange), 10 experts with $m = 2$, $k = 2$ (green), and 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameters and similar k/E . The dashed line at 1.0 marks monosemantic representation.

where $\|W\|_F^2$ denotes the Frobenius norm and p_e is the expected probability that expert e is used across a batch, equivalently the average renormalized gating weight after top- k routing.

Under equal total parameter budgets, the dense model has more features per dimension than the MoEs (Figure 3), and therefore relies more heavily on superposition. The dense model represents more learned features than total dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to remain near the monosemantic capacity limit ($n_{\text{learned}} \approx m_{\text{total}}$). As the total number of experts increases while total parameters and the ratio k/E are held roughly fixed, features per dimension decreases: more experts produce less superposition and more monosemantic feature representations. The dense model’s additional superposition also gives it consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with differences of $\sim 0.03 - 0.08$ at a given sparsity level. However, as expert count increases, MoE losses become consistently comparable to the dense baseline, indicating that increased network sparsity can improve interpretability while narrowing the reconstruction-loss gap. Appendix A.2 provides theoretical intuition for this pattern.

5 PHASE CHANGE

The preceding results show that MoEs and dense models can learn comparable numbers of features while producing markedly different interference structures. This motivates a finer analysis of whether the input-distribution variables that govern dense-model superposition—feature sparsity and feature importance—also govern MoE representations. We therefore examine whether these variables induce physics-inspired *phase changes* in expert representations, or whether network sparsity changes the nature of the transition.

Given finite representational capacity, a model can ignore a feature, represent it in superposition, or represent it monosemantically. A phase change occurs when the model switches discretely between these regimes across boundaries determined by feature sparsity and relative feature importance.

Dense toy models exhibit discontinuous ‘phase changes’ among internal feature representations (Elhage et al., 2022). Varying feature sparsity and relative importance can therefore elicit qualitatively different dense-model strategies; for example, higher feature sparsity encourages greater superposition. Expert-specific phase diagrams reveal that MoEs do not simply reproduce these dense-model regimes inside each expert, but instead adopt distinct allocation strategies under routing.

Setup. We use the experimental setup from Section 4.1, adding load-balancing loss in this section to avoid specialization collapse, in which routers and experts converge to local minima where some experts are ignored entirely. We evaluate three architectures, each with one active expert ($k = 1$): (A) $n=2$, $m=1$; (B) $n=3$, $m=1$; and (C) $n=3$, $m=2$. For each architecture, we report expert-specific phase diagrams over feature sparsity and last-feature relative importance while varying network sparsity by increasing the number of experts (E) up to the number of input feature dimensions (n).

In this section, we fix active rather than total parameters, setting $m_{dense} = km_{moe}$ and ignoring router parameters. This design isolates within-architecture representational behavior; otherwise, differences could be attributed to changes in active capacity rather than to routing and network sparsity. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. The latter, however, has only one hidden dimension ($m = 1$) for the same number of input features ($n = 3$), whereas the former has two hidden dimensions ($m = 2$), so the architectures impose different constraints on how superposition can express specialization.

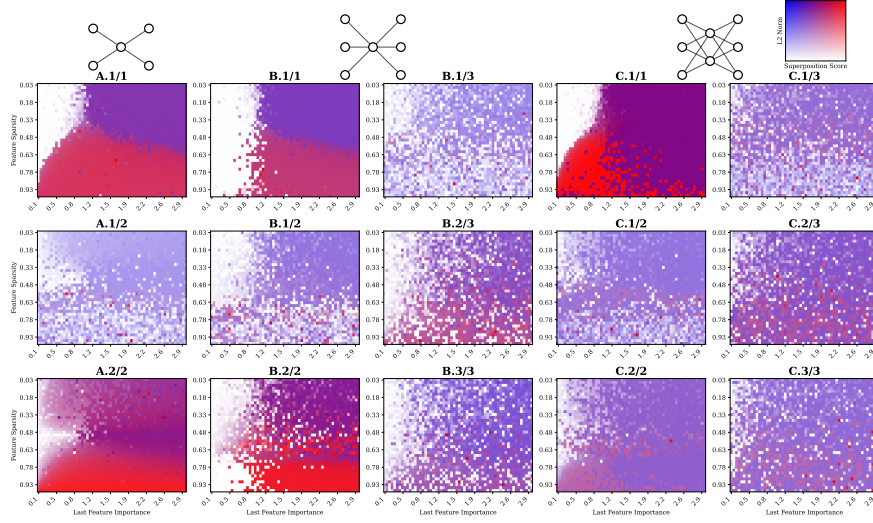


Figure 4: Expert-specific representation regime for the final input feature. For a given expert and feature, color decodes whether the feature is ignored (white), represented monosemantically (blue-purple), or represented in superposition (red). Each panel plots joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j<n} (\hat{W}_n \cdot W_j)^2$) over feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where subscript n denotes the final feature among n total features. For each cell, we train ten models and report the lowest-loss model, using load-balancing loss throughout this section. Low L2 norm ($\|W_n\|$) is shown in white, indicating that the model ignores the final feature; otherwise low superposition is shown in blue-purple, indicating monosemantic representation, and high superposition is shown in red. Cell (i, j) in subfigure X.e/E denotes expert e of E total experts trained on architecture X at last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the corresponding dense model.

Results and Takeaways. The single-expert dense cases exhibit clear phase changes (Figure 4.X.1/1), consistent with Elhage et al. (2022). Increasing the number of experts removes these sharp boundaries: MoEs with $E = 2$ sometimes resemble the corresponding dense diagrams (Figure 4.X.2/2), but their transitions are visibly more continuous. In each setting, at least one expert becomes more monosemantic and specializes in the most important feature by relative importance. Experts that diverge from the dense pattern have uniformly lower superposition scores, shown by bluer regions, and therefore cleaner feature representations. These phase diagrams reinforce the previous section’s conclusion that MoEs systematically favor lower superposition than matched dense counterparts.

In the $n = 2, m = 1$ architecture (Figure 4.A), the dense model ignores the final feature at low feature sparsity, whereas the comparable MoE preserves that feature more strongly because routing provides additional allocation flexibility. With three input dimensions (Figure 4.B), this preservation does not occur: experts superimpose the other two features, leaving insufficient one-dimensional capacity for the third feature. Unlike the other cases, architecture B lacks enough hidden capacity, even with superposition in the high-sparsity regime, to represent all features. Nevertheless, it still does not exhibit a clean dense-style phase change, except in the feature-sparsity window from approximately 0.5 to 0.7 in 4.B.2/2, where the transition is mostly discrete but mixed. For $m = 2$, the dense model’s white region (Figure 4.C.1/1) shows that it ignores the final feature in the mid- to low-feature-sparsity regime when that feature has lower relative importance. As network sparsity increases across the remaining panels in Figure 4.C, the final feature is represented with greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Thus, dimensionality in the low-relative-importance region increases with network sparsity, matching the aggregate trend in Figure 3.

Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 contain a feature-sparsity window from roughly 0.48 to 0.7 in which polysemantic representation, monosemantic representation, and feature ignorance coexist. This mixed region suggests that MoEs can encounter a middle regime where polysemantic and monosemantic solutions have comparable loss, making the optimizer less consistently committed to the strategies observed at low and high feature sparsity. Dense models do not display the same pattern, which further indicates that routing changes the representational landscape rather than merely reproducing dense phase behavior inside experts.

Conclusion. All experiments in this section use top- $k = 1$ routing, so only one expert is active for each input. Even under this restrictive condition, MoEs behave differently from dense models as soon as $E = 2$, including cases where hidden capacity with superposition is sufficient to represent all features. It is therefore mechanistically misleading to regard an MoE as a simple aggregation of dense models. The router restricts each expert to a subset of the feature domain, and that restriction substantially changes both expert learning and the geometry of feature representation.

6 EXPERT SPECIALIZATION

Having established that MoEs reduce superposition, we next ask how the resulting monosemantic features are organized within experts and how that organization should be understood as specialization.

Expert specialization in MoEs is often assessed through load balancing across all inputs (Chaudhari et al., 2025). Load balancing is important for utilization, but it does not capture the representational intuition behind specialization: an expert should be preferentially used when the concepts it represents are present in the input.

We therefore define an expert as specialized when it both *occupies* particular feature directions in input space and represents those features relatively monosemantically. We show that these two conditions are correlated in the toy setting, and that this feature-level specialization can also support balanced expert usage.

Because we fix $k = 1$, the router partitions feature space into convex cone regions (Appendix A.3), each assigned to a particular expert. By definition, if C is such a cone, then $\forall s > 0$, $\vec{x} \in C \rightarrow s\vec{x} \in C$, where s is any positive scalar. Thus, if a feature direction $\vec{x}$ is routed to expert e , every positive rescaling $s\vec{x}$ is routed to the same expert. We say that feature x is contained in expert e ’s routing region, and hence that expert e *occupies* x .

Empirically, small models that distribute input space across more experts tend to achieve lower loss (Appendix A.4). This observation raises a more mechanistic question: when the router allocates a region of input space to an expert, does that allocation predict what the expert represents? Our definition predicts that occupation of feature directions should be associated with *monosemanticity*, and the larger toy models below (e.g., $m = 10$) support this correlation.

For models with $n > 2$, we evaluate both directions of this relationship. First, we test whether initializing experts to occupy particular input-space features makes those experts more monosemantic with respect to those features. Second, for features that an expert ultimately represents monosemantically, we test whether the router also assigns those feature directions to the same expert.

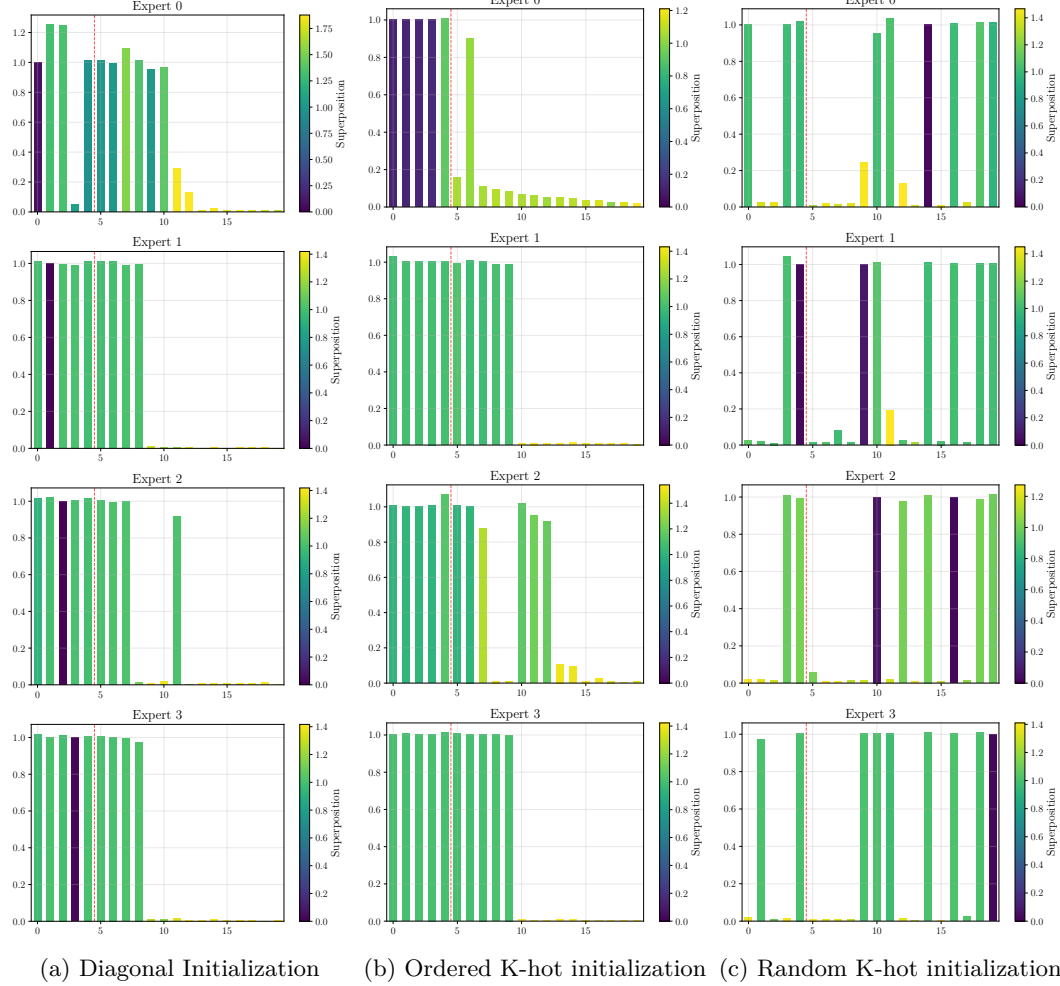


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition scores (color) under three initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially from feature one to feature 20. In (b), the gate matrix uses an “ordered k-hot” initialization: the first expert aligns with the first five features, and each subsequent expert aligns with the next five features. Relative feature importance matches (a). In (c), the gate matrix uses a “random k-hot” initialization, assigning each expert five random features so that experts share no feature while collectively covering all 20 features. Relative feature importance decreases exponentially but is randomly assigned across features.

With diagonal gate initialization, each expert monosemantically represents exactly the single feature it initially occupied (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of its five initially occupied features (Figure 5b). The remaining experts, initialized over less important features, do not monosemantically represent those features, nor do they instead capture the five most important features outside their initial routing region. When feature-importance ordering is broken through random k-hot

initialization, each expert monosemantically represents only the most important feature or features among those it initially occupied (Figure 5c).

These patterns show a strong correspondence between the features initially routed to an expert and the features that expert later represents monosemantically. They also show that importance mediates this correspondence: experts monosemantically represent important features, whether those features are assigned individually at initialization or included within a larger initialized feature set from which the expert selects the most important features.

With uniform feature importance, experts place all features in superposition and exhibit higher interference overall. Even in this less differentiated setting, initially occupied features remain *relatively* more monosemantic: their superposition scores are, on average, 1.03 standard deviations below the corresponding expert’s mean.

We next test the converse relationship: whether experts that represent particular features monosemantically also occupy those features in input space. We measure expert usage when the relevant features are *one of many* active features and when they are the *only* active features. The latter condition directly measures whether the expert occupies those feature directions. As Table 1 shows, the correlation holds under both xavier and k-hot initialization. With $E = 10$, mean usage of $\sim 10\%$ corresponds to balanced expert utilization; nevertheless, expert usage increases substantially whenever the expert’s monosemantic feature or features are active. When those features are the only active features, the corresponding expert dominates routing. Under k-hot initialization, 100% of all features represented monosemantically by an expert are also occupied by that expert.

Table 1: Monosemantic feature counts and routing-usage statistics per expert for $n = 100, m = 10, E = 10$. We train one hundred models for each initialization scheme (xavier and k-hot), yielding 1000 experts per scheme. Statistics are aggregated across models after grouping experts by the number of features they represent monosemantically. For each expert’s monosemantic feature(s), we report expert usage when those feature(s) are among several active input features and when those feature(s) are the *only* active input features.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

Experts that represent more features monosemantically are more specialized in the feature-level sense defined here. Their unconditional usage can be lower, but their conditional usage rises sharply when their specialized features are active. This pattern holds across the table except for the xavier-initialized case with a four-feature monosemantic expert, where utilization drops substantially.

7 CONCLUSION

We have shown that MoE routing changes the geometry of superposition rather than merely instantiating dense-model behavior inside separate experts. In controlled toy autoencoders, MoEs exhibit greater monosemanticity than dense networks, suppress dense-style phase changes, and localize interference within routed expert subspaces. We also introduced a feature-based definition of expert specialization and showed that, when initialization aligns experts with feature directions, experts organize around coherent monosemantic feature combinations. These findings are limited to synthetic reconstruction tasks and therefore leave open the extent to which the same mechanisms appear in large-scale transformers with unknown feature distributions (Appendix A.6). Within this setting, however, MoEs can approach dense-model reconstruction loss while maintaining more interpretable representations, challenging the view that mechanistic interpretability and model capability must be fundamentally opposed. Future work should determine which architectural and optimization conditions favor monosemanticity in MoEs, how MoE training dynamics diverge from dense-model dynamics, and when specialization emerges at scale.

REFERENCES

- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Marmik Chaudhari, Idhant Gulati, Nishkal Hundia, Pranav Karra, and Shivam Raval. Moe lens - an expert is all you need. In *Sparsity in LLMs (SLLM): Deep Dive into Mixture of Experts, Quantization, Hardware, and Inference*, 2025. URL <https://openreview.net/forum?id=GS4WXncwSF>.
- Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, and Aaron Sidford. Uniform sampling for matrix approximation. *CoRR*, abs/1408.5099, 2014. URL <http://arxiv.org/abs/1408.5099>.
- Nishanth Dikkala, Nikhil Ghosh, Raghu Meka, Rina Panigrahy, Nikhil Vyas, and Xin Wang. On the benefits of learning to route in mixture-of-experts models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9376–9396, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.583. URL <https://aclanthology.org/2023.emnlp-main.583>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Pub*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Google DeepMind. Gemini 2.5 pro. <https://deepmind.google/technologies/gemini/pro/>, 2025.
- Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *arXiv preprint arXiv:2305.01610*, 2023.
- Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.

-
- Jakub Krajewski, Jan Ludziejewski, Kamil Adamczewski, Maciej Pióro, Michał Krutul, Szymon Antoniak, Kamil Ciebiera, Krystian Król, Tomasz Odrzygóźdź, Piotr Sankowski, Marek Cygan, and Sebastian Jaszczur. Scaling laws for fine-grained mixture of experts. *arXiv preprint arXiv:2402.07871*, 2024. URL <https://arxiv.org/abs/2402.07871>.
- Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, Zhifeng Chen, Yonghui Wu, et al. Gshard: Scaling giant models with conditional computation and automatic sharding. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR, 2021.
- Siyuan Mu and Sen Lin. A comprehensive survey of mixture-of-experts: Algorithms, theory, and applications, 03 2025.
- Jungwoo Park, Ahn Young Jin, Kee-Eung Kim, and Jaewoo Kang. Monet: Mixture of monosemantic experts for transformers. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=10gw1SHY3p>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models, 2024. URL <https://arxiv.org/abs/2311.03658>.
- Lucas Prieto, Edward Stevinson, Melih Barsbey, Tolga Birdal, and Pedro AM Mediano. Correlations in the data lead to semantically rich feature geometry under superposition. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
- Adam Scherlis, Kshitij Sachan, Adam S. Jermyn, Joe Benton, and Buck Shlegeris. Polysemaniticity and capacity in neural networks, 2025. URL <https://arxiv.org/abs/2210.01892>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017. URL <https://arxiv.org/abs/1701.06538>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.
- Xingyi Yang, Constantin Venhoff, Ashkan Khakzar, Christian Schroeder de Witt, Puneet K. Dokania, Adel Bibi, and Philip Torr. Mixture of experts made intrinsically interpretable. In *Forty-second International Conference on Machine Learning*, 2025b. URL <https://openreview.net/forum?id=6QERrXMLP2>.
- Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Y Zhao, Andrew M. Dai, Zhifeng Chen, Quoc V Le, and James Laudon. Mixture-of-experts with expert choice routing. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho (eds.), *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=jdJo1HIVinI>.
- Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam M. Shazeer, and William Fedus. St-moe: Designing stable and transferable sparse expert models, 2022. URL <https://api.semanticscholar.org/CorpusID:248496391>.

A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

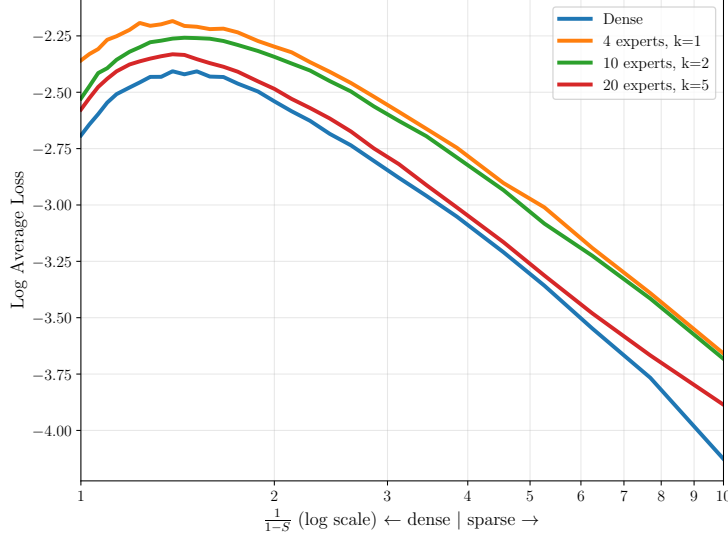


Figure 6: Log average loss versus feature density ($\frac{1}{1-s}$) for dense ($m = 20$), MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models with uniform feature importance ($I_i = 1.0$) and $n = 100$ input features. Results are averaged over five runs per sparsity level. The dense model has lower loss than all MoEs at every sparsity level, but increasing the number of experts consistently narrows the MoE–dense loss gap.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, however, “eat this same gap” structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs remove the computational penalty for having dedicated, rarely-activating neurons. As noted by Elhage et al. (2022), when a model only expends computation on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy, effectively trading the *compression* of superposition for the *selection* of routing.

The structural change manifests geometrically as a reduction in *interference*. While the global ratio of number of features represented to parameters remains constant, the router effectively partitions the feature space, ensuring that a feature routed to expert e only competes for capacity with the subset of features also assigned to that expert. Consequently, interference is governed by the local expert matrix $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning drastically reduces the number of interfering features for any given feature vector, minimizing the optimization pressure to pack features in superposition and allowing experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime of $k = 1$, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$. The region routed to expert i can be represented as $\forall j \neq i, (w_i - w_j)^\top x > 0$ where w_i and w_j are row vectors of W^r . This is a homogeneous linear inequality. Regions bounded by such inequalities are by definition convex cones. If a particular x satisfies this inequality, then multiplying both sides by any positive scalar s will still satisfy the inequality.

ity. Furthermore, if x_1 and x_2 satisfy this inequality, then any $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$ will also satisfy the inequality.

In the case of $k > 1$, the region of inputs which get sent to a particular expert e becomes a union of convex cones. Generally, the union of a convex cone is not itself a convex cone. Therefore, the understanding of experts occupying feature directions may not hold beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

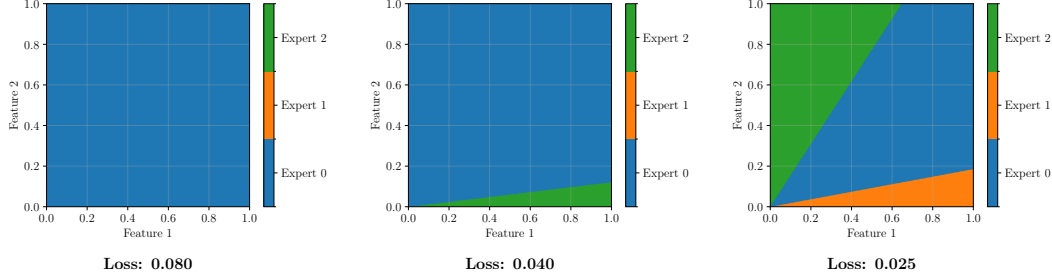


Figure 7: Expert routing of three identical models with differing initialization schemes. We use $n = 2$, $E = 3$, $m = 1$. The first model (left) has the worst performance (loss: 0.08) and routes all inputs to one expert. The second model (middle) has better performance (loss: 0.04) and routes a small portion of inputs, specifically those when feature 1 is active, to a second expert. The third model (right) has the lowest loss (loss: 0.025), and distributes the input space among all experts. One expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2$, $m = 1$, $E > 1$), we empirically find that models that distribute the input space across more experts tend to achieve lower loss, testing with $E \in [2, 7]$. Holding $n = 2$ allows us to visualize which portions of the input space get routed to which experts, as seen in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map between any dense model and a monosemantic MoE with an equivalent number of active features under a sparsity constraint.

Assume there exists an upper bound for the number of active features a for any input such that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Furthermore, assume that a is no greater than the hidden dimensionality, m , of an expert, providing an upper bound on the number of features a model has to represent. Assume also that the hidden dimensions is smaller than the total number of input features n ($a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —meaning the size of the subset of active features is smaller than or equal to a —create an expert which monosemantically preserves those features. (In fact, you can take only the subsets such that $|S| = a$.) The router then selects the expert which corresponds to those active features (of which there will never be more than a , by assumption):

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has sufficient capacity to represent its assigned features without superposition. To reiterate, only a features are active and every unique combination of active features receives its own dedicated expert with sufficient capacity to represent those features monosemantically. So, the number of possible experts needed is $\binom{n}{m}$.

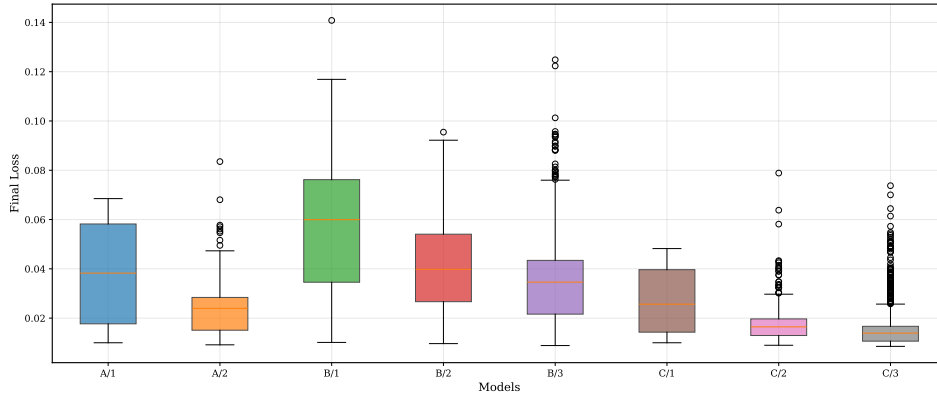


Figure 8: Model X/E uses X to denote the same model architectures and models used in Figure 4 and E denotes the total number of experts (i.e. network sparsity). Increasing network sparsity decreases mean loss while increasing localized variance—especially as the number of experts reaches the input feature dimensions. This can be attributed to the relatively unstable training of MoEs compared to dense models (despite training ten models for each cell and selecting the lowest loss).

The reconstruction for this theoretical MoE has zero loss only as the sparsity constraint holds (or goes to one in these toy models) because there is the chance more than m features could be active at one time ($a \not\leq m$), which would exceed the monosemantic representational capacity of the network (but the dense polysemantic could do no better unless features are correlated in the distribution). Therefore, even if $a \leq m$ sometimes, the polysemantic model encounters the same problem and the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

Thus, for any dense model, $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings described in this paper, the sparsity constraint holds in the limit where sparsity goes to one. However, in practice there may be an upper bound on the amount of features a particular amount of information can semantically encode, indicated by the size of meaningful embeddings of that data. Therefore, a MoE model with sufficient experts and a tractable amount of superposition (e.g. interpretable) may be sufficient to encode all features present.

A.6 LIMITATIONS

Our results should be interpreted in light of several limitations. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). While this enables precise measurement of interference and monosemanticity, it abstracts away many complexities of large-scale transformers, including multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics. Consequently, the transferability of our findings to real-world MoE architectures remains uncertain.

Second, our architectural and routing choices are intentionally restricted: we use simple experts, fixed top- k routing (often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs often employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022) or expanding beyond the task of reconstruction to next token prediction, for example, which may yield qualitatively different representational behaviors.

Third, the feature distributions in our synthetic tasks—including sparsity patterns, feature importance, and independence assumptions—are significantly simpler than those found in natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work suggests such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, our study provides mechanistic insights under clean experimental conditions, but further work is required to validate these patterns in large, realistic MoE systems. Unfortunately, our methods do not scale naturally to larger models and this is a clear direction for future work in this space.